

Introduction to Cyber Security

Cybersecurity is the process of protecting networks, systems, applications and digital assets from unauthorized access, cyberattacks, malware and data breaches. It ensures the confidentiality, integrity and availability (CIA) of information while supporting secure and sustainable digital infrastructure.

- Protects sensitive data using encryption, authentication and access control.
- Secures networks and systems against malware, ransomware and cyber threats.
- Supports safe digital transformation through secure cloud and IT infrastructure.
- Reduces downtime, resource misuse and operational risks caused by attacks.
- Enhances the reliability and resilience of modern digital technologies.

Core Pillars of Cybersecurity

Cybersecurity is based on three core principles that help protect information and systems.

CIA

- **Confidentiality:** Ensures that sensitive information is accessible only to authorized users.
- **Integrity:** Protects data from unauthorized modification and maintains its accuracy.
- **Availability:** Ensures systems, services and data remain accessible whenever needed.

Types of Cybersecurity

There are seven types of cybersecurity.

Types of Cybersecurity

1. **Network Security:** Protects computer networks from unauthorized access, malware and cyberattacks using firewalls, VPNs, intrusion detection systems and network monitoring tools.

2. **Application Security:** Focuses on securing software and mobile applications through secure coding practices, regular updates, vulnerability testing and security patches.
3. **Information or Data Security:** Protects sensitive data from unauthorized access, modification or loss using encryption, access controls, data classification and backup strategies.
4. **Cloud Security:** Secures cloud-based data, applications and infrastructure through identity management, encryption, compliance controls and secure configurations.
5. **Endpoint Security:** Protects devices such as computers, smartphones, servers and IoT devices using antivirus software, device encryption and regular system updates.
6. **Operational Security:** Includes security policies, access management, risk assessment, employee awareness programs and incident response procedures to protect organizational assets.
7. **Internet of Things (IoT) Security:** Protects connected devices such as smart home systems, wearable devices and industrial sensors through authentication, encryption and regular firmware updates.

Major Cybersecurity Threats & Attacks

Cybersecurity threats target systems, networks and data using various attack techniques. Some of the most common threats include:

Cyber Attacks

- **Malware Attacks:** Malware is harmful software such as viruses, trojans, ransomware and spyware that can damage systems or steal data
- **Phishing & Spear Phishing:** These attacks use fake emails, messages or websites to trick users into revealing sensitive information.
- **Ransomware Attacks:** Ransomware encrypts files and demands payment to restore access, often targeting businesses and organizations.
- **DDoS Attacks:** Distributed Denial-of-Service (DDoS) attacks overwhelm servers with excessive traffic, making services unavailable.

Emerging Trends in Cybersecurity

Cybersecurity is constantly evolving to address new threats and technologies. Some key trends include:

- **Rise of AI and Machine Learning:** AI-powered tools help detect, analyze and respond to cyber threats more quickly and accurately.
- **Rising Ransomware Attacks:** Ransomware attacks continue to grow, making regular backups and strong security measures essential.
- **Cloud Security:** As cloud adoption increases, organizations focus on securing cloud data and applications.
- **IoT Security Challenges:** Connected devices create new security risks, requiring stronger protection and regular updates.
- **Zero Trust Security:** Zero Trust verifies every user and device before granting access, reducing security risks.

Challenges of Cybersecurity

- **Evolving Threats:** Attackers constantly create new methods. Keep systems updated and monitor regularly.
- **Limited Budgets:** Security tools are costly. Prioritize critical assets and use cost-effective solutions.
- **Insider Threats:** Misuse of access. Enforce strict access controls and monitor user activity.
- **Complex Technology:** Cloud, IoT and hybrid systems increase complexity. Simplify systems and conduct regular security audits.

Cybersecurity Threats in 2025

Due to the increase in multiple technologies, businesses are getting a lot of advantages but to increase in technologies is also leading to the increase in several **cyber threats** by various processes. **Cyber threats can have major impacts on businesses of all sizes including financial and reputational damage.** This impact can vary depending on the severity of the attacks. Individuals and businesses should have knowledge about these cybersecurity threats and how hackers misuse the information from several techniques.

What are Cybersecurity Threats?

Cybersecurity threats are actions carried out primarily by [hackers](#) or attackers with malicious intent, with the goal of stealing data, causing damage, or interfering with computer systems. **The main categories of cyber threats are-malware, injection attacks, social engineering, configuration mistakes, and so on.** Cybersecurity can be originated from multiple sources from hostile nation states to individual hackers or contractors who abuse their privilege to perform the harmful acts.

The Evolution of Cybersecurity Threats

Cyber attacks have developed a long way over the past few years, and nowadays they are much more sophisticated, automated, and untraceable. Conventional security mechanisms are struggling to keep pace since cyberattackers are employing [Artificial Intelligence \(AI\)](#), [Machine Learning \(ML\)](#), and automation to perform mass cyberattacks with little or no human interference.

In 2025, the propagation of cyber threats will be influenced by a number of significant factors:

- **AI-Driven Cyberattacks** – Cyber attackers are employing AI to automate phishing attacks, deepfakes scams, and [malware](#) generation, rendering conventional detection useless.
- **Geopolitical Cyber Warfare** – Nation-states will escalate state-sponsored attacks against key infrastructure, financial systems, and government institutions.
- **Cloud Vulnerabilities & Supply Chain Attacks** – As companies quickly move to cloud computing, the hackers will take advantage of third-party security vulnerabilities and misconfigured cloud environments to acquire unauthorized access.
- **IoT Growth & Intelligent Gadgets** – Estimated billions of connected IoT devices worldwide will provide easy targets for security weaknesses in intelligent homes, medical networks, and industrial IoT networks to [cyber-attackers](#).

Common Sources of Cybersecurity Threats

Cyber threats originate from multiple **malicious entities**, each with unique attack strategies and motives. The most **dangerous sources of cybersecurity threats** include:

1. Cybercriminal Organizations

- [Cybercriminal](#) are groups of cyber attackers working as enterprises, stealing money, spreading ransomware-as-a-service (RaaS), and carrying out large fraud campaigns.
- They use ransomware, business email compromise (BEC), and financial fraud against corporations and individuals.

2. Hacktivists

- [Hacktivist](#) organizations perform politically or socially driven attacks to disrupt governments, organizations, or individuals.
- They employ DDoS (Distributed Denial-of-Service) attacks, data exposure, and web defacement to pass their message.

3. State-Sponsored Attacks

- State-sponsored hackers perform surveillance, cyber war, and infrastructure disruption.
- Nation-state actors attack financial institutions, defense agencies, and technology companies to steal confidential information or disrupt core services.
- Examples include [Advanced Persistent Threats \(APTs\)](#) that are concealed for prolonged intelligence gathering.

4. Insider Threats

- Authorized employees, partners, or contractors who have access to the company's sensitive information may accidentally or deliberately expose confidential information.
- In this manner, insiders are able to sell credentials, exploit privileged access, or be social engineered into committing massive breaches.

Top Cybersecurity Threats in 2025

Multiple Cybersecurity threats are performed by hackers to harm the data of an organization or business. Some of the **most important cybersecurity threats** are mentioned below:

1. Phishing Attacks

[Phishing Attacks](#) are well-known cybersecurity threats that are targeted through digital messages and are transmitted among people who have less knowledge of clicking on an unknown link that can install any harmful data. Phishing attacks are mainly performed by clicking on **suspicious links**. These attacks enable the hackers to steal the **user's login, personal financial information, and credit card credentials**.

2. Social Engineering

[Social Engineering](#) is one of the popular cybersecurity threats which is mostly dependent on human errors rather than technical errors which makes these attacks more dangerous. In 2024, social engineering strategies were the key method for getting the employee's data and credentials. More than **75% of targeted cyberattacks** begin with an e-mail. **Phishing is one of the well-known causes of data breaches**.

3. SQL Injections

[SQL Injections](#) is another **famous cybersecurity threat** which is a type of **code-based vulnerability** that allows the attacker to read and also to access personal data from the database. Therefore the attacker can use the sensitive data from the database and further can use [SQL queries](#) to modify, update, add, or delete the records in a database. This sensitive information may include the **company's data, user lists, or customer details**.

4. Vulnerabilities in Cloud

Cloud vulnerabilities are increasing and are one of the popular cybersecurity threats. The IBM reports confirm that cloud vulnerabilities have increased **150% in the past five years**. Gartner cloud security is one of the fastest-growing technologies in recent years. Verizon's DBIR has found that more than **90% of the 29000 breaches** analyzed in the report were mainly caused by website application breaches.

5. IoT Attacks

The [Internet of Things\(IoT\) attacks](#) is another famous cybersecurity threat that mainly involves adding internet connectivity to a system of interrelated

computing devices, digital machines, and mechanical machines. It has been observed that more than **70% of households** consist of at least one smartphone in their house which results in attacks on smart or **Internet of Things (IoT) devices** with more than **1.8 billion breaches occurring between January and June 2024**. IoT connectivity has opened a world of vulnerabilities for hackers and the average smart device is attacked within 5 minutes of connecting to the internet.

6. Low Data Management

Data management is very important in businesses and it is not just keeping the storage and organization's systems clean but it is also putting things in place. The amount of data has been developed by the consumers which doubles every four years but more than half of that new data is never used or analyzed. **Therefore piles of surplus data lead to confusion which leaves the data vulnerable to cyber attacks.** The breaches that are caused by data handling mistakes can be as costly as higher-tech **cybersecurity attacks**.

7. DDos

Distributed denial of service attack(DDos) is another famous attack that is done to disrupt the normal traffic of a targeted server or the network. Therefore, DDoS attacks are generally carried out with the networks of **Internet-connected machines**. These networks consist of the computers and the other devices which are actually been infected with the malware by further allowing them to be controlled by a hacker or attacker.

8. Ransomware

Ransomware is the type of malware that locks and encrypts a victim's data, systems, or files rendering them unusable until the attackers receive a **ransom payment**. The ransomware attacks also cost the companies in the form of income lost while the hackers hold the systems access for ransom. Therefore the average length of system downtime after a ransomware attack is 21 days.

9. Mobile Device Attacks

Mobile device vulnerabilities have increased in remote works which led to an uptick of companies implementing the bring your own device policies. Hence the cybercriminals targeted **mobile device management systems** which were designed to allow companies to manage the company devices in such a way that it keeps corporate data secure. For example - During COVID-19 the use of mobile devices has increased, hence not only the remote users depend on the mobile but also the pandemic experts also encourage large-scale adoption of mobile wallets

and touchless payment technology in order to control the germs transmission. Therefore a large population of users represents the larger target for cybercriminals.

10. Third-Party Vulnerabilities

The **third-party breach** occurred at the beginning of 2021 when the hackers leaked personal data from more than **214 million Instagram, LinkedIn, and Facebook accounts**. Hackers or attackers get around security systems by hacking the less protected networks that belong to the third party that has privileged access to the hacker's primary target. Therefore the hackers were able to access the data by breaching third-party contractors known as **SocialArks** that had been employed by the three companies and had privileged access to their networks.

Cybersecurity Challenges

Cybersecurity is crucial for businesses to thrive in today's competitive landscape. **Small and medium-sized businesses (SMBs)** particularly stand to benefit by safeguarding their reputation and protecting against **cyber threats**.

1. Third Parties Can Unlawfully Misuse the Potential of 5G Network

The advent of **5G networks** has sparked curiosity among the youth, offering the promise of more efficient use of their beloved gadgets. However, it also brings forth a challenge: the risk of emotional or physical attacks on this tech-savvy generation. Cyber assailants may unlawfully penetrate the **5G wireless networks**, exploiting the data collected or stored by smart and fast gadgets. These attackers, often third parties, capitalize on the complex architectures of 5G networks to carry out their malicious activities, leveraging revolutionary marketing tactics.

2. An Increasing Rate of Mobile Malware

Mobile malware is harmful software that intentionally targets the operating systems of mobile devices, disrupting their performance. The main reason for its

occurrence is the insecure usage of URLs over Wi-Fi or other internet networks. According to the **Mobile Security Report**, 97% of organizations face threats related to mobile malware from various vendors claiming to offer advanced security for cellular networks. Ignoring such vendors is not an option as they may introduce Trojan activities, cyber risks, and vulnerabilities.

The increasing rate of malware attacks on existing mobile phones has become a concerning theme during the COVID-19 pandemic. Various packages, such as tousanticovid.apk, covid.apk, covidMappia_v1.0.3.apk, covidMapv8.1.7.apk, and coviddetect.apk, are hidden in banking applications. When these applications are downloaded from malicious websites or associated hyperlinks, they expose mobile users to spam and other cybersecurity attacks. With the shift towards remote work, cybercriminals are expected to intensify their malware attack campaigns, making it crucial for mobile users to remain vigilant.

3. Artificial Intelligence: AI is Somewhere Controlling Cybersecurity Systems

Nowadays, healthcare industries and supply chain departments are adopting tools that support [Artificial Intelligence](#). Also, those tools have some glimpses of Machine Learning and NLP with which they are helpful in controlling the datasets primarily involved with patients' info or orders in which retailers/distributors are interested.

As per the McKinsey report, more than **25 percent** of healthcare organizations are investing in AI tools in this COVID-age. Even the banking sector has an impact of more than **30 percent** of the analytics derived via AI/ML tools.

The main loophole in using those Artificial Intelligent tools is that passwords and biometric logins are modified frequently by the patients, distributors, and other participants of the supply chain. With that, hackers can feasibly pick the pain points thereby controlling the monitoring of details like address, bank details, etc.

Since AI tools perform at minimal human input in real-times, healthcare and supply chain industries are sensing attacks of malware, ransomware strongly destroying their incentivize growth. No doubt, cybercriminals will be involved with data violence so that they can continuously gain access to that sensitive data for targeting more patients or supply chain participants.

4. Growing Popularity of IoT Devices

The usage of [Internet-of-Things devices](#) is trending nowadays because of their robust reaction-time and the lesser cost they invite in processing the merits of the cloud technology. Furthermore, the solutions those devices push through their

communication channels are incredible and considered by organizations comprising a varying number of workforces. However, with such growing popularity, cybercrimes are increasing continuously.

This is because cybercriminals can expose the profitable assets whose data is accessed from some industrial cloud network. In 2021, the IoT market has reached the potential of 418 billion US dollars, and we may expect it to grow to around **1.567 trillion USD by the end of 2025**.

All this is known to professional attackers primarily involved with selling the stolen data or acquiring control over the expanding operations of the businesses. With no hesitations, hackers are outwardly weaponizing the growing IoT popularity by destroying the channel nodes inviting prosperity plus the legitimate sales traffic for the organizations.

In this process, the protocols of cybersecurity maintaining and protecting the valuable data of customers have now become prominent to vulnerability.

5. Ransomware Attacks are Targeting the Critical Business Aspects

Ransomware attacks pose unpredictable threats to businesses of all sizes, impacting both **small and medium enterprises** and larger organizations alike, regardless of their familiarity with **data security** and compliance standards. According to Check Point Research, global ransomware attacks surged by 102% in 2021, with our country experiencing the brunt of 213 attacks weekly.

During these attacks, cybercriminals deploy **malware** or **viruses** targeting **mobile phones** and cellular networks, infecting connected devices like smartphones and laptops. This breach allows assailants to access personal information, leaving victims vulnerable to **extortion demands**. With over 1,000 organizations falling victim to **ransomware attacks** weekly, businesses must fortify their cybersecurity measures and safeguard against online criminal activities targeting their operations

6. No Control Over Phishing and Spear-Phishing Attacks

Spear-phishing attacks will easily be understood once we understand what phishing attacks are basically? So, phishing is somewhere related to social media and the cybercriminals prefer those phishing attacks because this helps them gather your card details (credit/debit), current location, or other sensitive info. Such attackers use deceptive emails or websites and show them in such a manner they look legitimate. Spear-phishing, on the other hand, is a sub-part of phishing and is its more sophisticated version. Here, online fraudulent send malicious emails, and they are sent to well-researched victims (such victims are

analyzed well by the cyberattackers on the grounds of mental and emotional strengths).

The **frequency of percentage** will vary **industry-by-industry** but the thing which will be common is sending **malicious emails** and attempting to access **personal data** through spammed websites. Besides, there are some **social media cybersecurity challenges** like acquiring control over the customers' accounts, **phishing** various campaigns running on social platforms like **Facebook, Twitter**, and misuse of data which is potentially important. All around the world, people are highly engaged with such **social media channels** primarily demanding internet services. This has made our privacy vulnerable to **phishers** or **spear phishers**, and they can confidently plan a series of events in hacking or destroying our personal wealth.

With those events, **hackers** would be navigating various sections of our personal **Facebook or Twitter accounts** and take advantage of such demonstrated weakness. And all this has created a sound disturbance in the security of **social media infrastructure**. Depending upon the scope of disturbance, [phishing/spear-phishing](#) attacks offer destruction to the privacy of user's data and the cybersecurity models of businesses too. Therefore, the issues, caused by destruction like **third party social media operators** supporting the tactics of **cybercriminals**, no close inspection of phishing emails at the users' accounts, violation of the right to information since users aren't aware of how their security is unknowingly compromised, are challenging the **cybersecurity protocols** of the organizations which need to be modeled with proper control and strengthened compliance standards.

7. Growth of Hacktivism

Hacktivism is a combination of words Hack N Activism. In general, this is done with the purpose of **breaking into someone's computer and steal that information that supports political or social agendas in the wrong way**. The target of hacktivists is primarily to gain their visibility on the websites of government organizations and deface their security protocols by promoting their politically influenced cause. According to the IBM X-Force report, there was **25 percent of data thefts and leak attacks** in which hacktivists have demonstrated their interest in seeking data of multi-national corporations and the government bodies connected with them. No matter what the intention of the hacktivists was, but such criminal attacks are a slap to government organizations taking care of the assets of their customers.

With this, a sort of motivation for challenging governments or forcing them to go against their morals is unknowingly promoted. There are many **anonymous hacktivist groups** working (since 2008) against disturbing the internal business processes of government or multinational organizations in the name of public welfare. They mix with the C-Level executives and continue embarrassing the government through the ideology of taking revenge with their online campaigns supporting regular flow [DDoS attacks](#). This is a newer version of breaking into cybersecurity systems of the government so that the protests of hacktivists may spread throughout the world and launch a shuttle of defacement of the reputation immorally.

8. Dronejacking is a New Wave Disturbing Cyber Experts

Dronejacking is a method cybercriminals use to exploit toy-like drones, gaining control over personal information easily. According to an Intel report, drones have targeted deliveries, camera crews, and hobbyists, bypassing enforced security standards. While drones are valuable tools for farmers, photographers, shippers, and law enforcement agencies, they also pose a new wave of cyber threats. **Dronejacking** enables cybercriminals to potentially cause financial harm to companies like **Amazon** and **UPS**, known for supplying essentials. Hackers can use **dronejacking** to determine package delivery schedules, posing a serious threat to consumer success and brand reputation.

Apart from financial implications, **dronejacking** presents other risks, such as the loss of expensive drones and damage to private property like commercial airplanes. This allows hackers to gauge response times and capabilities of drone controllers. Without staying updated on the latest security software and protection solutions, commercial operators and cybersecurity teams of larger organizations remain vulnerable to **drone attacks** and easy targets for anonymous criminals.

9. Preventive measures of social engineering

[Social engineering](#) is concerned with a type of cyberattacks where hackers focus on tricks and non-tech strategies rather than using core tech approaches or tools to trap the users. There are some preventive measures associated, and they are setting the spam filters from low to high, instant denial or deletion of help requests, researching the sources of unsolicited emails, and many more.

However, hackers are sophisticated nowadays and understand the frequency with which we are adopting such measures. They can feasibly take the legitimate access to our personal info and then, exploit us really well on the grounds of personality weaknesses. As per the report of Google, most of the **SEAs or Social**

Engineering attacks are phishing via official emails or malicious websites which almost look authentic.

In this graph, there are a number of accounts **up to 5000** flagged by Google, and they are trapped by the phishing attacks initiated by government-backed attackers. The number of attacks seems to fluctuate, yet it is clear that such SEAs are tracking our communications done through instant messaging or video conferencing. Furthermore, many of the knowledge-based workers, business owners, artisans (the number is **near to 260 million**) are remotely working in this pandemic era and this has made them vulnerable to such Social Engineering Attacks. Those cyber-attackers can smartly deploy multiple tactics for entering into their sensitive information like passwords, usernames, and banking details. All this will look legitimate as they will be using trademarks, logos of the well-known companies whose **accuracy will be 99 percent** or a little higher.

As soon as their websites and emails are clicked, you will be tricked and then, the height of your awareness is primarily destroyed. Furthermore, the organizations are also prone to such steals as hackers are easily breaking their business software backed by systems adhering to cybersecurity protocols. So, the security of their infrastructure is inclined towards the likelihood of success of these cybercriminals well-versed with how they should be using the preventive measures of social engineering for deploying their malicious agendas?

10. Office People Having Access to Data of their Organizations

Internal politics is something that everyone is aware of and this happens in every organization. Whether you talk about a tech-giant or a well-reputed automation agency, employees are assigned with some privileges and this makes the finances vulnerable to huge losses. All this gives rise to **insider threats**. They have grown up by 47 percent in the past 2 years and successfully inviting cybercriminals to nourish their fraudulent activities well.

More than **34 percent** of businesses are affected every year by such threats and this is giving the way to accidental breaches for breaking the trust and reputation of customers. Those **insider threats** are underestimated by the businesses a lot as they think it is important for them to deal with the complex market trends rather than giving such threats a look! All this disturbs the current status of a company as their employees have signed some deals with hackers for providing them the important information about the company.

Cybersecurity Solutions

With **cybersecurity threats in 2025** becoming more **sophisticated**, organizations must implement **advanced security solutions** to mitigate risks. Below are key strategies to protect against modern cyberattacks.

1. Preventing Phishing Attacks

- Apply AI-based email security to identify phishing attacks.
- Apply [Multi-Factor Authentication \(MFA\)](#) to protect accounts.
- Apply social engineering awareness training for employees.

2. Stopping Social Engineering Attacks

- Train employees on actual social engineering techniques.
- Enforce [Zero Trust Security](#), which verifies all users prior to providing access.
- Utilize behavioral analytics to identify abnormal user behavior.

3. Defending Against SQL Injection & Injection Attacks

- Sanitize user inputs and use **parameterized queries**.
- Deploy [Web Application Firewalls \(WAFs\)](#) to block malicious [SQL commands](#).
- Conduct regular **penetration testing** to identify vulnerabilities.

4. Securing Cloud Environments

- Implement **cloud security solutions** with real-time threat detection.
- Enable **encryption and identity management** for sensitive cloud data.
- Use **Cloud Access Security Brokers (CASBs)** to monitor cloud activity.

5. Mitigating IoT Attacks

- Deploy IoT security frameworks to safeguard connected devices.
- Deploy AI-based security solutions to track in real-time.
- Segment IoT networks to restrict lateral movement in the event of an attack.

6. Strengthening Data Management Practices

- Implement data loss prevention (DLP) policies.
- Implement periodic data backup and recovery solutions.
- Implement strict access control policies to limit unauthorized access to data.

7. Preventing DDoS Attacks

- Use AI-based DDoS protection software to detect malicious bursts of traffic.
- Use [intrusion prevention systems \(IPS\)](#) to block evil traffic.
- Implement rate limiting and network segmentation to secure.